BIOL-8 Exam 02 — Answer Key

Modules 07-10: Genetics, Cell Division, Tissues & Inheritance

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Part A: Multiple Choice (30 points)

Q#	Answer	Explanation
1	B	Central dogma: DNA → RNA → Protein
2	C	A codon = 3 bases (triplet code)
3	B	Transcription produces mRNA from a DNA template
4	B	A DNA mutation sometimes changes the protein (silent mutations don't, missense mutations do)
5	B	Semiconservative: each new molecule has one original + one new strand
6	B	Mitosis = growth and repair of body cells
7	B	DNA replication occurs in S (Synthesis) phase
8	C	Mitosis produces 2 genetically identical diploid cells
9	C	After meiosis: 23 chromosomes (haploid) in each sex cell
10	D	Meiosis produces 4 haploid cells
11	C	Muscle tissue specializes in contraction
12	C	The digestive system breaks down food into absorbable nutrients
13	B	The stomach wall contains muscle tissue (smooth muscle for churning food)
14	D	Neurons are the functional cells of nervous tissue
15	B	Skeletal, cardiac, and smooth are the three muscle types
16	C	The circulatory system transports oxygen and nutrients via the blood
17	C	Skeletal muscle is the only type under voluntary (conscious) control
18	C	Aa × Aa genotypic ratio = 1 AA : 2 Aa : 1 aa (1:2:1)

Q#	Answer	Explanation
19	B	1/4 (25%) will be aa (recessive phenotype)
20	B	Red × White → all pink = incomplete dominance (blending in heterozygote)
21	C	ABO blood type: multiple alleles (I^A, I^B, i) and codominance
22	C	Carrier = heterozygous, shows dominant phenotype, carries recessive allele
23	C	A dihybrid cross (heterozygous × heterozygous for 2 traits) yields 4 phenotype combinations
24	B	Unaffected parents → affected child = autosomal recessive
25	B	Males (XY) have one X; one recessive allele is enough to cause the condition
26	C	50% of sons receive X^c from carrier mother
27	B	Genes on different chromosomes sort independently during meiosis
28	D	AO × BO → possible genotypes: AB, AO, BO, OO (all four blood types)
29	B	Autosomal recessive: requires two copies (one from each parent) to express the trait
30	B	Mothers can pass X-linked recessive alleles to sons via their X chromosome

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Part B: Fill in the Blank (11 points)

Q#	Answer	Notes
1	diploid	Two complete sets of chromosomes (2n)
2	gametes	Sperm and egg cells
3	haploid	One set of chromosomes (n)
4	mitosis	Cell division for growth and repair
5	meiosis	Cell division that produces sex cells
6	epithelial	Covers surfaces and lines cavities
7	connective	Blood, bone, cartilage, adipose all belong to this category
8	muscle	Specialized for contraction
9	genotype	Genetic makeup (letters like Aa)
10	phenotype	Observable traits (what you see)
11	dominant	Masks the expression of the recessive allele

Part C: Free Response (9 points)

Students choose 3 of 5 questions, 3 points each.

1. Mitosis vs. Meiosis

- Mitosis: growth and repair; produces 2 diploid cells; genetically identical to parent cell (1 pt)
- Meiosis: produces sex cells (gametes); produces 4 haploid cells; genetically varied (1 pt)
- Clear comparison showing understanding of purpose, number, and ploidy (1 pt)

2. Blood Type Genetics

- Correct Punnett square: $I^Ai \times I^Bi$ (1 pt)
- Four offspring genotypes: I^AI^B, I^Ai, I^Bi, ii (1 pt)
- Four blood types identified: AB, A, B, O — each 25% (1 pt)

3. Tissues and Function

- First tissue: names specific example + explains structure-function link (1.5 pts)
- Second tissue: names specific example + explains structure-function link (1.5 pts)
- Example strong answers:
- Epithelial → simple squamous in alveoli: single thin layer for rapid gas exchange
- Connective → blood: liquid matrix (plasma) allows transport through vessels
- Muscle → cardiac: striated with intercalated discs for coordinated heartbeat
- Nervous → neurons: long axons transmit electrical signals over distance

4. From DNA to Protein

- Transcription: DNA → mRNA in the nucleus (1 pt)
- Translation: mRNA → protein at ribosomes (in cytoplasm or on rough ER) (1 pt)
- Accurate naming of steps and locations; clear description of information flow (1 pt)

5. X-linked Inheritance

- Mother is a carrier (one normal X, one affected X); father is unaffected (X Y) (1 pt)
- Sons get their only X from mom → 50% chance of receiving the affected allele → one allele is enough to cause the condition (1 pt)
- Daughters get an X from each parent → would need two copies of the recessive allele; carrier mom + unaffected dad means daughters can be carriers but not affected (1 pt)

End of Answer Key